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Decentralized Certificate Authentication and Fraud Prevention Using Blockchain for Educational Institutions and Employers

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***Abstract*—The increasing issue of fake educational and professional certificates are creating serious difficulties in the verification of certificates. The proposed project will focus on implementing a decentralized certificate verification and fraud protection scheme based on concept of blockchain technology, with the intention of achieving secure, transparent, and tamper-proof digital certificate verification. The proposed project will apply cryptographic hash functions to produce digital signatures of certificates, unchangeable on the blockchain. The proposed system has three primary modules: a certificate issuance and blockchain registration module for the administrator, a verification module for organizations, and a scanning module that scans certifications using quick response technology to make the real time verification possible. By using the quick response code from the certificate and associating it with blockchain-based information, there is greater usefulness in terms of transferability. The system identifies alterations made to certificate using a hash function.**

***Index Terms*—Blockchain technology, certificate authentication, fraud prevention, decentralized systems, digital credentials**

I. Introduction

The issuance and verification of academic and professional certificates is really crucial in the education and employment sector and with the huge surge of certificate forgery, tampering, data manipulation and single point failures in the pre-existing manual systems, it is really crucial to use a better robust framework. These limitations reduce trust for academic credentials and increase administrative overhead to institutions and employers [1]–[3].

We have designed a blockchain-based decentralized certification issuance and verification system that ensures security, transparency, and real-time validation to tackle the aforementioned issues. During certificate issuance, an unique digital identifier is created for the certificate data using cryptographic hash functions, which was then stored on a blockchain record [4], [5]. All these transactions would be immutable meaning that any changes to the certificate would result in a hash mismatch, immediately indicating that the certificate has been tampered [5], [6].

In order to ensure an efficient verification the certificate holder is issued a QR code generated from a blockchain-stored hash. To verify the certificate either the QR code is scanned or submitting the verification string through a web interface, thus allowing thus third parties to validate authenticity without relying on the issuing authority [7], [8]. With the use of blockchain's decentralized and tamper resistant nature, our system prevents fraudulent logins and significantly reduces verification time and increases transparency [9], [10].

II. Related Work

There has been multiple studies which has shown the extensive use of blockchain technology to build a secure interface for certificate issuance and verification. Some earlier blockchain-based research models demonstrated that decentralized records could effectively prevent unauthorized modification and trusted verification of certificate records without having direct involvement of issuing authority. In order to make sure that the setup was transparent and immutable it was integrated with smart contracts [1].

Later works relied on using cryptographic hash values and QR based authentication for making the verification mechanism more efficient [4]. These approaches were done by linking the blockchain-stored hash values to the physical or digital certificates enabling rapid verification even by third part institutions [2], [5]. While these systems were proven to be effective at the institutional level, it often faced interoperability and scalability issues when expanded to national or international systems due to regulatory and policy constraints [11]. The later works focused on improving accountability and transparency by allowing domain owners to directly control certificate lifecycles while allowing independent verification through public logs [6], [7]. This approach was done to address the limitations of traditional Public Key Infrastructure (PKI). These approaches may not be suitable for real-time or high-throughput environments as they might introduce extra computational overhead and latency during validation [8]. Although the use of blockchain and other encrypted

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applications in data security, many existing systems still rely on manual processes or centralized databases which hinders efficiency causing single-point-failures [3], [9], [12]. While the use of blockchain has increased a lot in financial and identity management systems, its usage in academic and professional credential verification remains limited [10], [13]. There is still a lack of practical, decentralized and tamper-free verification system which can be jointly adopted by both educational institutions and employers without complex mechanisms or high operational overhead [14], [15]. This work proposes a blockchain-based certificate verification framework that gives real-time validation which is decentralized and secure system through QR code-based verification. This framework is aimed to bridge the gap between theoretical blockchain models and deployable credential verification solutions particularly made for academic and professional environments.

III. Proposed Methodology

In this work, we designed and implemented a system which focuses on ensuring that the process for issuing and verifying a certificate is highly secure and reliable by using blockchain and QR code technologies. The process as described in figure 1 below begins with Education Authority acting as issuing authority for an academic certificate. After the student certificates are uploaded through a secured system by the Education Authority, each student certificate will then be assigned a unique QR code. This QR code acts as a cryptographic identifier which will link the physical certificate to a digital record that will be held on a blockchain. After the registration is completed, the certificate holder is then able to download and share their QR code for proof of legitimacy. The Education Authority also uploads information regarding certificates and has organized databases regarding the companies. Therefore, all documents will be organized in a way that communication between parties can occur easily rather than wasting time searching for documents in order to perform verification requests. They can also either provide the requesters with information on their specific academic past or update with every request. The system allows the requester or the employer to be able to check if the student is legitimate or not from the educational records as the databases are organised. This proposed system is completely interlocked and will help in improving trust and communication between parties with proper verification compared to manual verification and paperwork.

QR code verification is intended to be easy and convenient. In case of verification required, the verifier scans the code using a device with a webcam or QR reader. The process is started by running a pre-configured batch file that opens the webcam. After scanning, the QR code is checked against the hash stored on the blockchain. When the QR code is valid and is a match for the corresponding record, the student's certificate information is fetched and displayed on the screen, validating the authenticity of the certificate. When the QR code is invalid or does not exist in the system, it displays an error message stating a certification attempt failure.

Admin Module: The admin represents the educational authority responsible for managing certificate records. Once the admin logs into their account using their respective login username and password they can then upload the details of the student along with their corresponding . These records are then stored securely on the blockchain, where each certificate is assigned a unique hash value, referred to as a digital signature. A QR code is generated based on this hash value, which can be provided along with the certificate. When scanned using a mobile device, the QR code retrieves certificate details directly from the blockchain. If the QR code data matches with what is stored on the blockchain, the certificate is deemed authentic.

Company Module: This part of the proposed work comprises of the employers or institutions allowed to register and log in to the system. Once authentication has been carried out, users can upload or scan a candidate certificate. The system then generates a digital signature from the uploaded certificate and compares with the existing signatures stored on the blockchain. If the system finds a match, it authorizes the certificate authenticity, confirming that it has not been fake or reformed.

Scanner Module: The dedicated standalone module called scanner module is maintained by educational institutions or companies. This module allows the users to scan QR codes allied to certificates, retrieve the associated data from the blockchain, and verify the document legitimacy. This module confirms a rapid and secure method of certificate validation. To employ Blockchain for storage requires SOLIDITY contract which has roles to STORE and authenticate certificate details in it. This solidity contract needs to be installed on Ethereum Blockchain. However it returns contract deployed ADDRESS, this address can be specified in PYTHON code to store and access certificate details.

A. Algorithm

This section delves on the steps involved in the proposed blockchain-based framework for secure certificate issuance and verification. These algorithms make sure that the model is secure, integrated and transparent throughout.

Algorithm 1: Blockchain Certificate Issuance

Require: Admin Credentials C_{admin}, Student Details S, Certificate Document D

Ensure: Transaction Hash T_x, Verification QR Code Q

1) $Auth \leftarrow AdminLogin(C_{admin})$
2) if $Auth$ is Valid then
3) $H_c \leftarrow SHA\text{-}256(D, S)$ ▷ Generate unique hash
4) $B_i \leftarrow \{S_{id}, H_c, \textit{Timestamp}\}$ ▷ Prepare ledger record
5) $T_x \leftarrow RecordTransaction(B_i)$ ▷ Push to blockchain
6) $AwaitConfirmation(T_x)$ ▷ Ensure persistence
7) $Q \leftarrow EncodeQR(H_c)$
8) return T_x, Q

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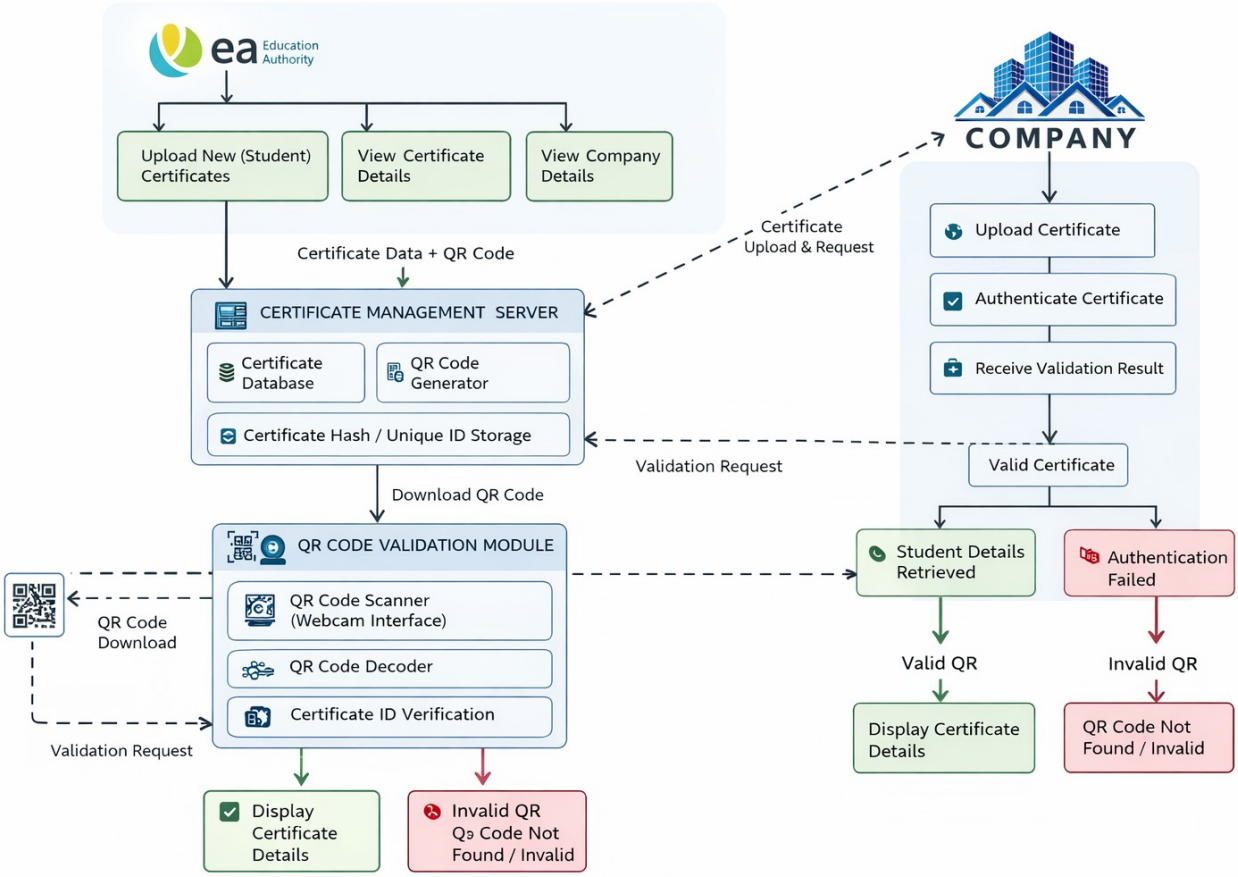


Fig. 1. Proposed system architecture

9) else
10) return $Error$("Authentication Failed")
11) end if

Algorithm 2: Certificate Verification Process

Require: QR Code Q or Certificate Document D
Ensure: Validity status $V \in \{\text{Valid}, \text{Invalid}\}$

1) if Input is Q then
2) $H_{query} \leftarrow DecodeQR(Q)$
3) else
4) $H_{query} \leftarrow SHA\text{-}256(D)$
5) end if
6) $Record \leftarrow QueryBlockchain(H_{query})$ ▷ Check ledger for hash H_c
7) if $Record \neq \emptyset$ then
8) $V \leftarrow Valid$
9) else
10) $V \leftarrow Invalid$ ▷ Possible forgery detected
11) end if
12) return V

IV. Implementation and Results

The demonstration of the proposed system used three main modules Admin, Company, and Scanner. In the Admin module the educational institutions or approved staff are allowed to log in to the system and submit the details of the student and documents presenting the certificate. Once the submission is done by the admin, the system then creates a digital hash of the uploaded certificate using hashing algorithms, which will then be put into an immutable part of the blockchain. A corresponding QR code was then be created and attached to the certificate as a physical link to the subsequent digital verification process.

The Company module was intended for use by employers or third-party verifiers. It allowed intuitive scanning of a candidate's certificate QR code. Once scanned, the system inspected the scanned hash against the blockchain record. If a match was found, the certificate was verified and the verifying details were retrieved and displayed. If the certificate was absent from the blockchain or it was authenticated but tampered with, the verification failed, immediately alerting the verifier to the possibility of forgery.

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TABLE I
EXPERIMENTAL SETUP

Parameter	Description
Blockchain	Ethereum (Local test network – Ganache)
Smart Contract Language	Solidity
Backend	Python (Web3.py)
Hash Function	SHA-256
QR Generation	Python QR library
System Modules	Admin, Company, Scanner
Test Certificates	650 academic certificates

Moreover, the Scanner module improved the verification process by allowing direct verification using mobile devices. This module would be very helpful during campus placements, hiring events, or academic admissions. The QR code-based mechanism, paired with blockchain verification, enabled quick and real-time verification, all requiring minimal resources.

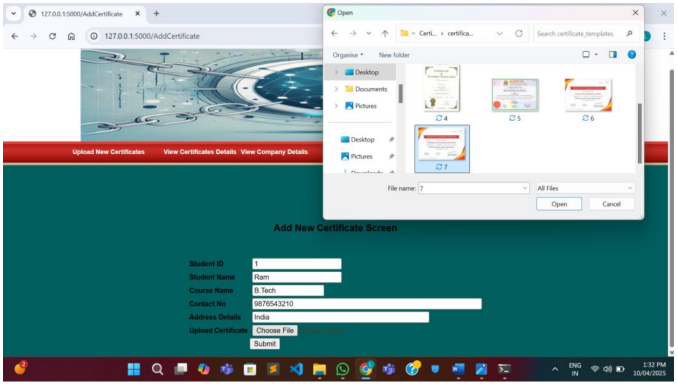


Fig. 2. Certificate upload page

The above Figure 2 displays the certificate upload section, where the admin can upload certificates and enter the details of the certificate receiver.

The demonstration validated that blockchain technology, if implemented properly, could revolutionize the way credentials are managed. The way proposed work handles credentials/higher education certificates has been plagued with issues such as the need for physical verification, slow background checks, and certificates manipulated incorrectly. The demonstration validated the design goals of the system- security, transparency, trust, and ease of use. It allows for expansion in the ways that could improve; i.e. biometric validation, extending this to non-academic official documents, or linking to national repositories. Admin Login page allows administrators to securely log in using username and password.

The above Figure 3 shows the screen displayed after a certificate is uploaded. It contains all the uploaded certificate details along with a system-generated hash code and a unique QR code for verification purposes. The Company Signup screen in their respective platform allows companies to register by providing necessary details such as company name, password, contact number and email ID, enabling them



Fig. 3. Uploaded certificate details page

to authenticate certificates in the platform. The company uploads a certificate for authentication purposes. This feature allows companies to verify the authenticity of certificates by submitting them to the system for validation.

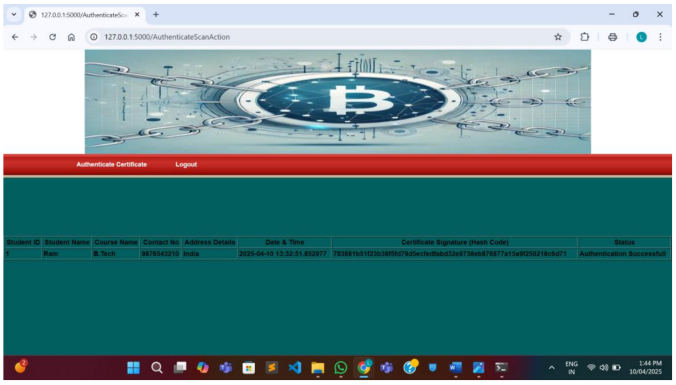


Fig. 4. Certificate authentication details

The screen displayed after a certificate is uploaded and successfully validated is shown in Figure 4. It displays all the certificate details including certificate receiver's details uploaded date and time, and unique hash code, confirming its authenticity and allowing the company to review the verified information.

When no matching QR code is found during scanning, it displays the message "Certificate verification failed. QR code does not exist", indicating that the scanned certificate is not recognized or is invalid. The blockchain-based digital certificate verification system demonstrated strong performance in terms of security, efficiency, and reliability. By addressing the growing problem of forged academic credentials, the system leveraged the Ethereum blockchain and Solidity-based smart contracts to ensure the integrity and immutability of certificate data. During issuance, a cryptographic hash representing the certificate content was generated and permanently stored on the blockchain. A QR code derived from this hash enabled

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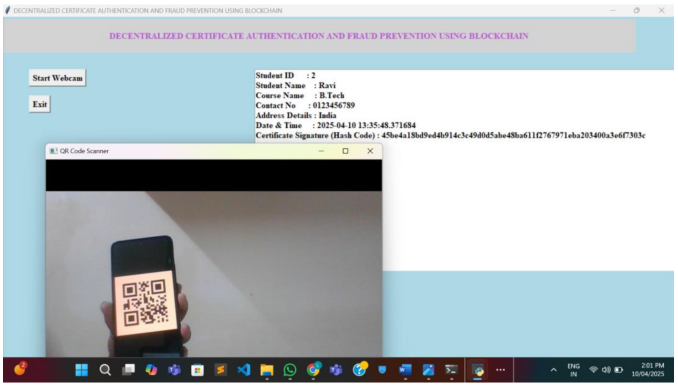


Fig. 5. Successful certificate authentication using QR code

rapid verification. Testing confirmed that valid certificates were authenticated instantly, while forged or altered documents failed verification due to hash mismatches. This process eliminated reliance on centralized authorities and enabled third-party verification through web and mobile interfaces. The decentralized architecture ensured fault tolerance and data availability even when network nodes were offline. The system also demonstrated scalability by handling multiple concurrent verification requests with consistent performance. Furthermore, the user-friendly interface allowed non-technical stakeholders, such as administrators and recruiters, to interact with the system without understanding the underlying blockchain processes.

The system showed strong resistance to tamper, as any modification to certificate resulted in immediate detection through hash validation. While the proof of concept was implemented in a local Ethereum environment, the architecture can be extended to public blockchains and integrated with institutional and recruitment platforms. Overall, the results indicate that blockchain technology provides a viable, transparent, and secure solution for improving trust, verifiability, and governance in academic credential management. The proposed system was evaluated to test its efficiency in verification of the certificates, reliability and integrity of the system in contrast to the pre-existing approaches like manual verification and centralized database-based methods. The results from the experiment indicates that the proposed system significantly reduces certificate verification time by eliminating manual background checks and centralized dependency which is depicted in Table II. This is possible due to verification being done using QR code scanning and blockchain hash giving near real-time authentication compared to long delays in manual and centralized systems due to multiple intermediate steps. The unchanging nature of blockchain storage ensures that any small change to certificate data results in a hash mismatch, therefore immediately identifying forged or tampered certificates during verification.

Unlike centralized systems, which can be susceptible to insider attacks and single-point failures from within, the decentralized ledger allows independent verification by third-party entities without requiring direct interaction with the

TABLE II
PERFORMANCE EVALUATION OF PROPOSED SYSTEM

Metric	Manual	Centralized	Blockchain + QR
Avg. Verification Time (s)	120	45	5
Forgery Detection	Low	Moderate	Very High
Single Authority Dependency	High	High	None
Tamper Resistance	Low	Medium	Very High
Real-Time Verification	No	Limited	Full
System Reliability	Low	Medium	High

issuing institution. The experiment was carried out with 600+ certificates and the results obtained are depicted in Table III and Figure 6 and Figure 7. The system also effectively prevents unauthorized modification of certificate records, as all issued certificates are recorded on the blockchain with cryptographic hash values and thus improving security, transparency, building trust, and availability across academic and professional verification scenarios.

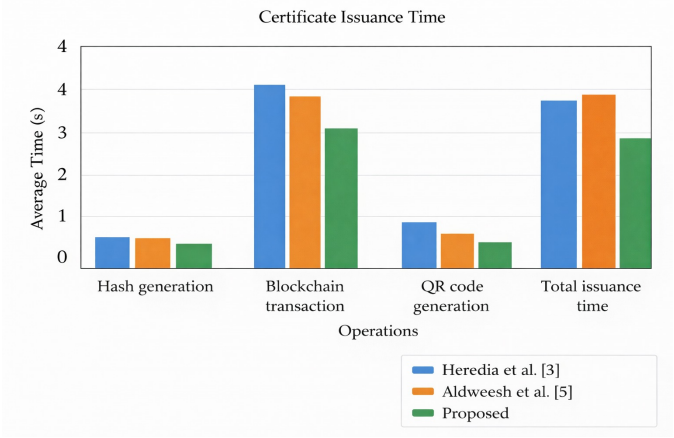


Fig. 6. Comparison of certificate issuance time with existing systems

Overall, the results demonstrate that the proposed system fulfils its objective of providing a secure, efficient, and tamper-resistant certificate verification mechanism suitable for educational institutions and employers.

TABLE III
CERTIFICATE ISSUANCE TIME (IN SECONDS)

Operation	Heredia et al. [3]	Aldweesh et al. [5]	Proposed
Hash generation	0.25	0.22	0.18
Blockchain transaction	3.2	2.8	2.4
QR code generation	0.24	0.18	0.12
Total issuance time	3.9	3.2	2.7

V. CONCLUSION

The rising increase in forged academic and professional certificates indicated of the faulty pre-existing manual and centralized verification systems. In this work, we have proposed a blockchain-based certificate verification framework to address the challenges faced by the traditional approaches as they are often time-consuming, prone to manipulation, and dependent

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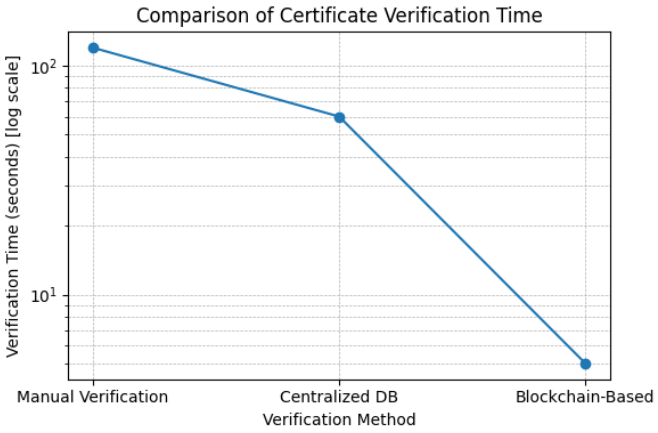


Fig. 7. Comparison of average certificate verification time

on a single verifying authority. These limitations make existing systems unfit for large-scale and real-time credential validation. The proposed system uses cryptographic hashing and immutable blockchain storage to securely record certificate information, enabling reliable detection of any unauthorized modification. By integrating QR code-based verification, the system significantly reduces verification time as the authenticity of the certificate could be verified without requiring direct interaction with the issuing authority and improving resistance to forgery and refining the reliability of the model.

Even with the additional overhead during the certificate registration, this overhead only occurs during the time of issuance and does not affect the verification process. In this proposed work, certificate validation is shown to be light and fast, thus making the trade-off acceptable for practical deployment. The results indicates that use of blockchain to solve the issues faced by traditional methods is practical and feasible in academic and professional environments. Future work may focus on integrating the system with national education repositories, optimizing transaction costs, and extending the framework to support other forms of official credentials.

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